

ABERDARE TECHNICAL COLLEGE

DEPARTMENT OF INFORMATION & COMMUNICATION TECHNOLOGY

UNIT: PERFORM PROGRAMMING PRINCIPLES

UNIT CODE: 0613 451 05A

CAT: 1 MARKING SCHEME

MODULE / TERM: MODULE 4, TERM 4 | MARKS: 50 |
TIME: 1.5 HRS

QUESTION 1 — Programming Languages & Categories (7 Marks)

a) Define the term 'programming language.'

A formal set of instructions, commands, symbols, and syntax used to write code that a computer can understand and execute to perform specific tasks. *(2 marks)*

b) Briefly describe three categories of programming languages, giving one example language for each:

- Procedural Languages: Structured around sequential procedures or routines (functions/subroutines) that operate on data step-by-step. Example: C, Pascal, COBOL. *(1.5 marks desc + 0.5 mark ex)*
- Object-Oriented Languages: Structured around 'objects' that bundle data (attributes) and code (methods) together to model real-world entities. Example: Java, C++, Python. *(1.5 marks desc + 0.5 mark ex)*
- Functional Languages: Based on mathematical functions that avoid changing state and mutable data (declarative style). Example: Haskell, Lisp, Scala. *(1.5 marks desc + 0.5 mark ex)*

QUESTION 2 — Low-Level vs High-Level Languages (4 Marks)

Aspect	Low-Level Languages	High-Level Languages
Abstraction	Machine-dependent with little/no abstraction from hardware.	Machine-independent with high abstraction from hardware.
Execution & Syntax	Executes very fast; cryptic binary or mnemonic assembly code.	Requires compilation/interpretation; human-readable English-like syntax.
Examples (1 mark ea)	Machine Code, Assembly Language (1 mark)	Python, Java, C++, C# (1 mark)

QUESTION 3 — Criteria for Language Selection (4 Marks)

State four criteria considered when selecting a programming language for user requirements:

- Type of Application/Domain (e.g., web development, system programming, data science). *(1 mark each - Any 4)*

- Performance and Speed Requirements (execution efficiency and memory consumption needs). *(1 mark each - Any 4)*
- Availability of Libraries and Frameworks (pre-built modules to speed up development). *(1 mark each - Any 4)*
- Developer Experience / Team Skillset (expertise of the available development team). *(1 mark each - Any 4)*
- Security Features (built-in safeguards for data protection and safe memory management). *(1 mark each - Any 4)*
- Platform Compatibility and Portability (ability to run on target operating systems/hardware). *(1 mark each - Any 4)*

QUESTION 4 — Scenario-Based Selection: Hospital System (10 Marks)

Identify and explain five criteria to select a programming language for a hospital management system (2 marks per point: 1 mark for explanation + 1 mark for direct hospital context):

- **Security and Data Protection:** Hospital systems store sensitive patient medical records and billing data subject to data privacy regulations (e.g., Kenya Data Protection Act). The language must support strong encryption, secure authentication, and robust memory safety. *(2 marks)*
- **Database Integration & Data Processing:** Managing records, appointments, and billing requires frequent CRUD operations. The language must have powerful database drivers and ORM frameworks (e.g., Java with JDBC/Hibernate, C# with Entity Framework). *(2 marks)*
- **System Reliability & High Availability:** Hospital operations run 24/7. System downtime or uncaught runtime exceptions can disrupt critical healthcare services; strong type-checking and exception handling are mandatory. *(2 marks)*
- **Scalability & Performance:** As patient traffic, record volumes, and concurrent user requests grow over time, the language and framework must scale effectively without severe performance degradation. *(2 marks)*
- **Integration Capabilities / Interoperability:** Must connect with external software/hardware (e.g., medical laboratory machines, insurance provider APIs, SMS notifications, and payment gateways like M-PESA). Requires RESTful API support. *(2 marks)*

QUESTION 5 — Programming Paradigm Definition (2 Marks)

a) Define the term 'programming paradigm.'

A fundamental style, pattern, or theoretical approach to structuring and designing computer programs, defining how computations are conceptualized and executed. *(2 marks)*

QUESTION 6 — Common Programming Paradigms (6 Marks)

Explain three common programming paradigms and their problem-solving approaches (2 marks each):

- **Imperative / Procedural Paradigm:** Focuses on HOW a task should be accomplished using a step-by-step sequence of explicit statements that change the program state over time. *(2 marks)*

- **Object-Oriented Paradigm (OOP):** Models problems around real-world entities (objects) containing data (attributes) and behavior (methods), solving problems by facilitating interaction between objects. *(2 marks)*
- **Declarative / Functional Paradigm:** Focuses on WHAT the program should accomplish rather than how to do it step-by-step, evaluating pure mathematical functions without mutating data. *(2 marks)*

QUESTION 7 — Functional Programming (4 Marks)

a) Define functional programming.

A declarative programming paradigm where computation is treated as the evaluation of pure mathematical functions, avoiding state changes and mutable data. *(2 marks)*

b) State two key characteristics of functional programming:

- **Immutability:** Variables and data structures cannot be modified once created. *(1 mark)*
- **Pure Functions:** Functions return identical outputs for identical inputs and produce no side effects. *(1 mark)*

QUESTION 8 — Procedural Programming (5 Marks)

a) Define procedural programming.

A programming paradigm derived from structured programming, based on the concept of procedure calls (routines, subroutines, or functions), executing a linear sequence of computational steps. *(3 marks)*

b) Give two advantages of using the procedural paradigm:

- **Simplicity and Ease of Understanding:** Straightforward linear logic flow, ideal for learning and simple scripting. *(1 mark)*
- **Code Reusability & Low Overhead:** Functions can be reused across programs with low memory consumption and fast execution. *(1 mark)*

QUESTION 9 — Object-Oriented Programming (4 Marks)

a) Define object-oriented programming.

A programming paradigm structured around software objects that combine state (data/fields) and behavior (code/methods) to simulate real-world concepts. *(2 marks)*

b) State two core principles of object-oriented programming:

- **Encapsulation:** Binding data and methods into a single unit (class) and restricting direct access. *(1 mark)*
- **Inheritance:** Allowing a child class to acquire properties and behaviors from an existing parent class. *(1 mark)*

QUESTION 10 — Imperative vs Declarative Paradigms (4 Marks)

Aspect	Imperative Paradigm	Declarative Paradigm
Core Focus	Focuses on HOW to solve a problem	Focuses on WHAT outcome is desired

	(step-by-step state changes).	without defining explicit control flow.
State Management	Relies heavily on mutable state, variables, and loops.	Minimizes or eliminates mutable state and side effects.
Example Language	C, Pascal, Java (1 mark)	SQL, Prolog, Haskell, HTML (1 mark)